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Tabbed: A Python package for reading variably structured text files at scale

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Summary

Delimiter separated value (DSV) text files are ubiquitous for representing tabular data. For example, a search of Github.com for comma separated value files, a subset of the DSV format, returns 75 million matches. They are simple to create, easy to share, and can encode a variety of data types. Despite their broad use, variability in the formatting and structure of DSV files has frustrated their automated parsing for decades. The *clevercsv* package (Burg et al., 2019) made significant progress on this problem with consistency measures that accurately detect a file's dialect: delimiter, quote character, and escape character. The structure of a DSV file introduces another source of variability. DSVs may contain a metadata section that offsets the header and start row. Tabbed uses row length and type consistency measures to automatically detect metadata, header and data sections of a DSV file. Furthermore, Tabbed provides a value-based conditional reader for reading these irregular DSV files at scale.

Statement of need

To the best of our knowledge, no parser of DSV files exists that can locate the start of the data section irrespective of the presence of metadata and/or a header. Further, we found no reader of DSV files that can conditionally read rows from the data section based on a row's type casted values. We here define a set of desiderata for parsing irregular text files that optionally contain metadata, header, and ragged data rows. To motivate this set, we consider the DSV shown in Figure 1.

```
1  Experiment ID; Experiment
2  Animal ID;  Animal
3  Researcher;  Test
4  path = '/home/user/experiments/states.txt'
5
6
7  Number | Start Time | Runtime | Annotation
8  0 | 02/09/22 09:17:38.948 | 0.0000 | Started Recording
9  1 | 02/09/22 09:37:00.000 | 1161.0520 | start
10 2 | 02/09/22 09:37:00.000 | 1161.0520 | exploring
11 3 | 02/09/22 09:37:08.784 | 1169.8360 | grooming
12 4 | 02/09/22 09:37:13.897 | 1174.9490 | exploring
```

Figure 1: The first 12 lines of a delimiter separated file named *sample.txt* with metadata on lines 1-4 and a header on line 7. The metadata contains both semicolon-delimited and undelimited strings. Notice the data section uses a different delimiter than the metadata and contains mixed data types.

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Structural Detection

A header line (line 7 of Figure 1) marks the boundary between the metadata and data sections of a file. Detection of this line is critical for correct automated parsing. Tabbed can locate a header line using string inconsistencies or type inconsistencies depending on the data types represented in the file. Importantly, some irregular DSV files do not have a header. In this case tabbed generates a header based on the number of data columns it measures from a sample of the file.

Type Casting

Strings in the data section of the file shown in Figure 1 represent mixed types that need to be type casted. Tabbed supports conversion to `int`, `float`, `complex`, `time`, `date` and `datetime` instances. These conversions are graceful, returning strings on failure and logging the conversion problem for post-reading introspection.

Value-based Filtering

Selectively reading rows from a DSV file based on type casted content is extremely useful for selecting subsets of the data. For example, in the sample file of Figure 1, users may want to only read data rows where the column named *Annotation* has a string value of *exploring*. Tabbed supports both column selection and row filtering with equality, membership, rich comparison, regular expression matching, and custom callables. To support an intuitive interface for creating these filters, tabbed uses simple keyword arguments passed to a method called 'tab' of the Reader class. Below we illustrate the simplicity of constructing these filters for the sample file shown in Figure 1.

```
from tabbed.reading import Reader

with open('sample.txt', 'r') as infile:
    reader = Reader(infile)
    # Tell the reader to only read these Tabbed rows
    reader.tab(Start_Time ='>=2/09/2022 9:37:00', Annotation = 'exploring')
    result = reader.read()
```

Iterative Reading

DSV files have no file size limits making it essential that readers support file streaming. Tabbed's reader returns an iterator whose per-iteration memory consumption is tunable. For speed, this feature is implemented using a first-in-first-out (FIFO) data structure with O(1) time complexity allowing tabbed to linearly scale to large files.

Comparison

Tablib (Reitz, 2016), comma (Lumbroso, 2020), and pandas (The pandas development team, n.d.) are popular alternative packages to tabbed. Table 1 compares the features of these packages. Pandas `read_csv` function most closely matches the available features in tabbed but it has two draw-backs; it does not detect where the data section begins, and it does not support conditional reading of data rows. The first issue is perhaps the most serious because it requires the user to tell pandas where to begin reading via the `start` parameter for each irregular DSV file.

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Table 1: Comparison of features for four common open-source software packages for reading DSV files. Plus (+) and minus (-) indicates package support or lack of support for each feature respectively.

Software	Structural Detection	Type Casting	Filtereable	Iterative
tablib	-	+	limited	-
comma	-	limited	-	-
pandas	-	+	columns only	+
tabbed	+	+	+	+

Given that pandas `read_csv` most closely matches with tabbed's capabilities, we tested tabbed's read speeds against pandas in Figure 2. For this test, we selected the `python` engine in pandas rather than the `c` or `pyarrow` engine for parsing files. This is important because these alternative engines achieve an order of magnitude greater read speed over the python engine. The speed test was conducted on a DSV file with all floats and a DSV file with floats and datetime instances. The file size in each case was 10 columns by 100,000 rows. The left panel of Figure 2 shows that tabbed is slower on both the float and mixed type files by approximately 30%. Nevertheless, users can expect to read millions of cells from a DSV file in just a few seconds with tabbed.

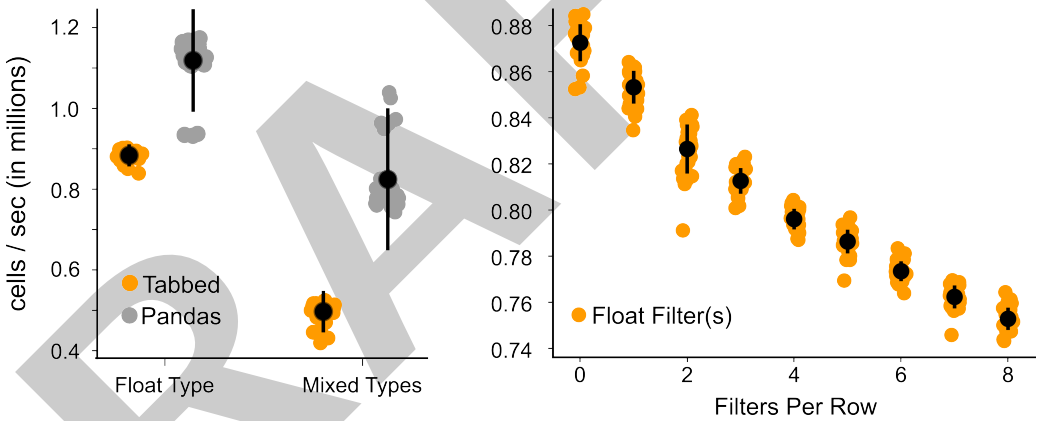

Figure 2: Tabbed and pandas read speed comparison. Left: Comparison of number of cells casted per second between tabbed and pandas for DSV files composed of floats or mixed types. The conversion engine for pandas was chosen to be "python". Right: Tabbed's read speed as a function of the number of filters applied to each row during reading. In both panels black circles and error bars are the mean and standard deviation across 30 trials. These comparisons were carried out on a single 2.4 GHz Intel Core i5-6300U processor.

To further understand tabbed's performance, we tested how row filtering impacts read speeds. The right panel of Figure 2 shows the read speed as a function of the number of filters applied to each data row. The left-most point is the baseline with no filters. We measure an approximately 2% reduction in speed for each new filter added.

Conclusion

Automated reading of large irregularly structured DSV files, a format that is broadly used, is an open challenge. Tabbed's four features—structural detection, type casting, value-based filtering, and iterative row reading—work together to automate reading of these files. The simple interface and deeply tested API of these features (see documentation) makes tabbed accessible to a broad audience of analysts, researchers, and developers across disciplines.

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